WHEN THERE IS NO RIGHT ANSWER: EVALUATION FAILURE AND A SAMPLING-SCALE FIX FOR LOW-RESOLUTION PIXEL-ART TEXTURES

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ABSTRACT

Low-resolution pixel-art texture generation exposes a blind spot in how generative models are evaluated. When the output is a 16×16 grid of discrete colour cells, independent artists drawing the *same* material agree on 9.8% of cells against 8.6% expected under a paired null (50,753 artist pairs) — there is no unique right answer. We show this single fact breaks both mainstream automatic evaluation routes. Per-pixel metrics measure marginal fit rather than quality. And VLM judges fail in ways that per-item agreement cannot detect: on the same human-labelled set, two judges agree with the human on *identical* fractions of items (66.7%), yet one compresses an 86% effect to 62% (p=0.14) and erases its stratification while the other preserves both; a third inverts the human conclusion outright. Under human blind comparison a trivial downsampling baseline beats a purpose-built masked-prediction model 83:17. The bottleneck is not model capacity but a *structural-unit convention* mismatch: the generator draws $\sim$25 units per tile regardless of canvas while artists draw $\sim$4.5, measured independently from 5,979 artist tiles and 125 artist sources. Cropping to the detected dominant period fixes it, and manipulating the unit count produces a monotone dose–response in the fix's benefit (88%, 54%, 17% at 27.7, 13.8 and 9.4 units). We pre-registered and *falsified* a pure sampling-limit account along the way; reporting that failure is what selects the surviving mechanism.

1 INTRODUCTION

[PENDING: prose; argument chain in docs/draft_en.md §1]

2 RELATED WORK

[PENDING: prose; four-way positioning in docs/related-work.md]

3 THERE IS NO RIGHT ANSWER

[PENDING: 3.1 agreement measurement, 3.2 per-pixel corollary, 3.3 judge failure]

4 A TRIVIAL BASELINE IS ENOUGH

[PENDING: 4.1 baseline vs artist, 4.2 baseline vs specialised model]

5 THE REAL BOTTLENECK: STRUCTURAL-UNIT CONVENTION

[PENDING: 5.1 diagnosis, 5.2 fix, 5.3 gate, 5.4 pre-registered falsification]

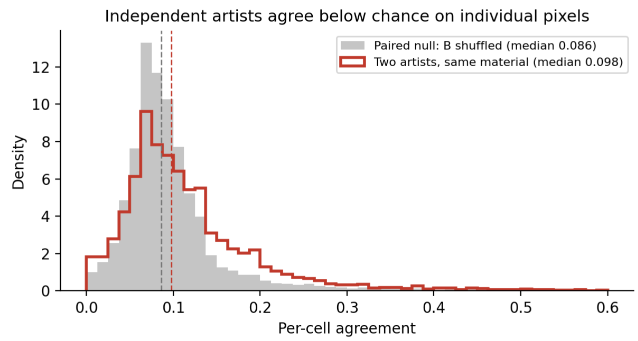

Figure 1: Independent artists agree on individual cells barely above a paired null. The null shuffles the second artist's own tile, preserving its marginal while destroying its arrangement, so both sides pair the same marginals.

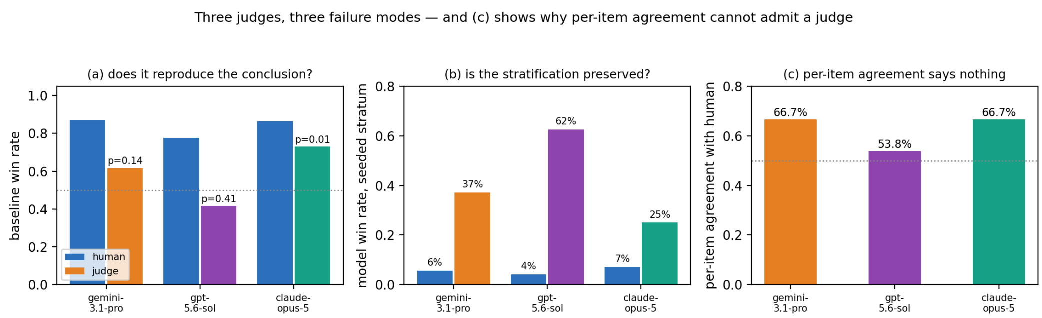

Figure 2: Three judges on the same human-labelled pairs. Panel (c) is the methodological point: per-item agreement — the quantity the VLM-as-judge literature reports — is identical for the judge that destroys the conclusion and the one that preserves it.

6 Limitations

[PENDING: single annotator; 4.1 power; conditional win rates; per-image gate does not follow from the condition-level effect]

Reproducibility Statement

Every claim in this paper is traceable to a numbered experiment record in the project log, including the six results we retracted. Pre-registrations are git commits timestamped before the corresponding runs.

References

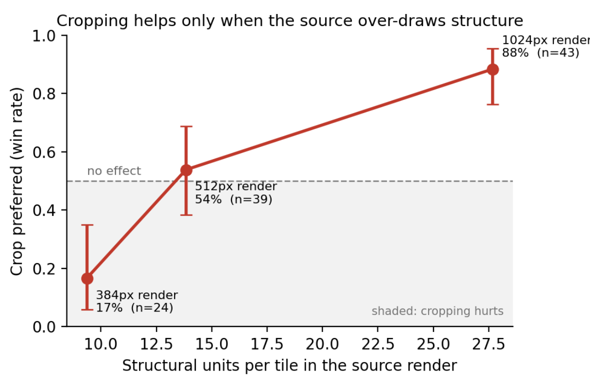

Figure 3: Manipulating how many structural units the generator draws produces a monotone change in whether cropping helps, with the midpoint sitting at no effect. Bars are Jeffreys intervals.